

3. What is the rate of fission energy release (MeV/sec) for a 5mg sample of pure U-235 if it were to be irradiated at an REP at the PULSTAR (thermal flux = $1\text{E}13$ n/cm²/sec, epithermal flux = $2\text{E}11$ n/cm²/sec). Mass of U-235 is 235.044 g/mol. Show your work, the nuclide block is provided.

1/+	U235	7/-
26 m IT ~76.8eV(ω)e ⁻	0.7204 7.04E8 a α 4.398, 4.366, ... γ 185.72, 143.76, ... SF $\nu\omega$ σ_f 98, 14E1 σ_f 585, ~275 $\sigma_a < 0.1$ mb 235.043930	

4. 6 Factor Formula Definitions - *MATCHING*

Factor:	Definitions:
_____ η	A. Ratio of the number of thermal neutrons that are absorbed by the fuel, to the number of thermal neutrons that are absorbed in the core.
_____ ϵ	B. Ratio of fast neutrons from both fast & thermal fission, to the number of fast neutrons from thermal fission alone.
_____ p	C. Ratio of the number of thermal neutrons that are absorbed in the core, to the number of neutrons that reach thermal energy.
_____ f	D. Ratio of the number of neutrons generated from thermal fission, to the number of thermal neutrons absorbed in the fuel.
_____ $\mathcal{L}_{\text{TH}}$	E. Ratio of number of fast neutrons that start to slow down (moderate), to the number of fast neutrons from fast and thermal fission.
_____ $\mathcal{L}_{\text{F}}$	F. Ratio of the number of neutrons that reach thermal energy, to the number of neutrons that started to slow down (moderate).

5. Fill in the blanks with either a value or a range of values where appropriate ($K_{\text{eff}} = 1.025$ or $0 < \rho \leq 390\text{pcm}$) using the correct units. (Hint: if $\rho = \$1 = 730\text{ pcm}$, then $K_{\text{eff}} = 1.0073$)

If the reactor is:	K_{eff} is:	Reactivity ρ (pcm) is:
Super-prompt critical:		
Prompt-critical:		
Super-delayed critical:		
Critical:		
Sub-critical:		

Walkdown Lab Questions:

1. PULSTAR Systems - **Matching** (*answers may be used more than once, or not at all...*)

- _____ 1. # Fuel Assemblies in PULSTAR Core
 _____ 2. # of fuel pins per fuel assembly
 _____ 3. # of neutron detectors (NI's) monitoring the reactor power
 _____ 4. # of **Scrammable** Control Rods
 _____ 5. Number of SCRAM interlocks
 _____ 6. # of Cooling Loop Pumps
 _____ 7. Year the PULSTAR went critical.

A.	25
B.	20
C.	14
D.	7
E.	4
F.	3
G.	2
H.	1
J.	1972

2. What are the three (3) main functions of the primary water?

3. What is the function of the three delay tanks in the PPV?

4. Which describes the correct flow path of heat from the reactor core to the atmosphere?

- a) Core, ¹⁶N Delay Tank, Primary Pump, Heat Exchanger, Cooling Tower
 b) Core, ¹⁶N Delay Tank, Heat Exchanger, Primary Pump, Cooling Tower
 c) Core, Primary Pump, ¹⁶N Delay Tank, Heat Exchanger, Cooling Tower
 d) Core, Heat Exchanger, ¹⁶N Delay Tank, Primary Pump, Cooling Tower

Startup Lab Questions:

- a) About how many times (generally) do we double the Neutron Flux Monitor (NFM) count rate to bring the reactor from shutdown to super-criticality? _____
 b) What type of neutron detector (hint: 'X'-chamber) is the NFM connected to? _____
 c) List two indications on the reactor console that you can use to determine if the reactor is supercritical:

 d) Number the order (step #1 through #6) of the INPO Guidelines for reactor operators:

Step #	INPO Guideline
	OBSERVE – The result of the control manipulation.
	VERIFY – Am I about to manipulate the correct control?
	TOUCH – Which control does the procedural step indicate should be manipulated?
	ANTICIPATE – How should the reactor systems respond to the manipulation?
	STOP & THINK – What operations evolution/procedure is being performed?
	MANIPULATE – Perform the control manipulation.

BONUS:

SCRAM Matching – match the SCRAM to the setpoint (*answers may be used more than once, or not at all...*)

<u>Ans.</u>	<u>SCRAM</u>	<u>Setpoint</u>	
_____	1. Linear Power Channel – Over Power SCRAM	A.	Red Button
_____	2. Safety Power Channel – Over Power SCRAM	B.	1.10 MW
_____	3. Log-N Channel – Flow/Flapper SCRAM Enable	C.	120°F
_____	4. Safety Power Channel – Flow/Flapper SCRAM Enable	D.	475 GPM
_____	5. T ₂ Pool Temp	F.	-36 inches
_____	6. Pool Level	G.	116°F
_____	7. Primary Coolant Flow	H.	1.20 MW
_____	8. Manual SCRAM	J.	150 kW

What thermocouple IDs are used to measure the temperature change across the core?

Provided Equations, Conversions, and Reminders:

$$\rho = (K_{\text{eff}} - 1) / K_{\text{eff}}, \text{ where } \rho \text{ is in units of } \frac{\Delta k}{k}$$

$$K_{\text{eff}} = 1 / (1 - \rho), \text{ where } \rho \text{ is in units of } \frac{\Delta k}{k}$$

$$\beta_{\text{eff}} = 730 \text{ pcm} = 0.0073 \frac{\Delta k}{k} \text{ (assumed value for PULSTAR)}$$

$$1 \text{ kg} = 2.21 \text{ lb}_m$$

$$1 \text{ MW} = 3.41 \times 10^6 \text{ BTU/hr}$$

$$N_A = 6.022 \times 10^{23} \text{ atoms/mol}$$

Scientific Notation Reminder: $1\text{E}15 = 1 \times 10^{15}$